9:56

vulms.vu.edu.pk/Qu

3

BIO302 - Molecular Biology (Quiz 2) Quiz Start

Question # 10 of 10 (Start time: 09:55:56 PM, 25 July 2023)

The diameter of exit tunnel of bacterial ribosome is about _____ at its narrow points

Select the correct option

- $3A^0$
- $10A^0$
- $5A^0$
- $1A^0$

Click to Save Answe

Solved by Muhammad Kashif(03064963933)

9:55

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3

BIO302 - Molecular Biology (Quiz 2) Quiz Start

Question # 9 of 10 (Start time: 09:55:39 PM, 25 July 2023)

_____ processes are carried out before mRNA is being exported to the cytoplasm from the nucleus?

Select the correct option

- ○ Transcription and Pre-mRNA splicing
- ◉ Transcription, Pre-mRNA splicing and Polyadenylation
- ○ Only Pre-mRNA splicing
- ○ Transcription and polyadenylation

Click to Save Answe

Solved by Muhammad Kashif(03064963933)

9:55

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BIO302 - Molecular Biology (Quiz 2)

Question # 8 of 10 (Start time: 09:55:25 PM, 25 July 2023)

The three nucleotide long units of protein coding region of the mRNA is calle

Select the correct option

○	Pre-codon
○	Tricodon
○	Anticodon
●	Codon

Solved by Muhammad Kashif(03064963933)

9:55

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BIO302 - Molecular Biology (Quiz 2)

Question # 7 of 10 (Start time: 09:54:59 PM, 25 July 2023)

In RNA editing which of the following enzyme is not involved?

Select the correct option

- ○ Terminal uridylyl transferase (TUTase)
- ○ Endonuclease
- ○ RNA ligase
- ◉ RNA polymerase

Solved by Muhammad Kashif(03064963933)

9:54

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3

BIO302 - Molecular Biology (Quiz 2)

Question # 6 of 10 (Start time: 09:54:25 PM, 25 July 2023)

Which initiation factor binds with small subunit and blocks it from re-associating with a large subunit?

Select the correct option

- [x] IF3
- [ ] IF2
- [ ] IF1
- [ ] IF4

Click to Sa

Solved by Muhammad Kashif(03064963933)

9:54 vulms.vu.edu.pk/Qu 3

BIO302 - Molecular Biology (Quiz 2)

Question # 5 of 10 (Start time: 09:54:13 PM, 25 July 2023)

One of the unusual bases found in tRNA is?

Select the correct option

○	Adenine
○	Guanine
●	Hypoxanthine
○	Cytosine

Solved by Muhammad Kashif(03064963933)

9:54

vulms.vu.edu.pk/Qu

3

BIO302 - Molecular Biology (Quiz 2) Quiz Start Time: 09:52 PM

Question # 4 of 10 (Start time: 09:53:38 PM, 25 July 2023)

The transcription machinery and the translation machinery in _____ are located in the same compartment.

Select the correct option

Reload Mc

- Viruses
- Eukaryotes
- Prokaryotes
- Fungi

Answer & Move to Ne

Solved by Muhammad Kashif(03064963933)

9:53

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3

BIO302 - Molecular Biology (Quiz 2) Quiz Sta

Question # 3 of 10 (Start time: 09:53:26 PM, 25 July 2023)

How many enzymatic steps are required by all aminoacyl-tRNA synthetases to attach an amino acid to a tRNA?

Select the correct option

○	1
●	2
○	3
○	4

Click to Save Answ

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9:53 vulms.vu.edu.pk/Qu 3

BIO302 - Molecular Biology (Quiz 2)

Question # 2 of 10 (Start time: 09:53:09 PM, 25 July 2023)

The anticodon loop contains _____.

Select the correct option

- A three nucleotide codon
- A four nucleotide anticodon
- A four nucleotide codon
- A three nucleotide anticodon

Solved by Muhammad Kashif(03064963933)

9:53

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3

BIO302 - Molecular Biology (Quiz 2)

Question # 1 of 10 (Start time: 09:52:33 PM, 25 July 2023)

The Amino acids polymerizing machinery is composed of at least _____ RNA molecules.

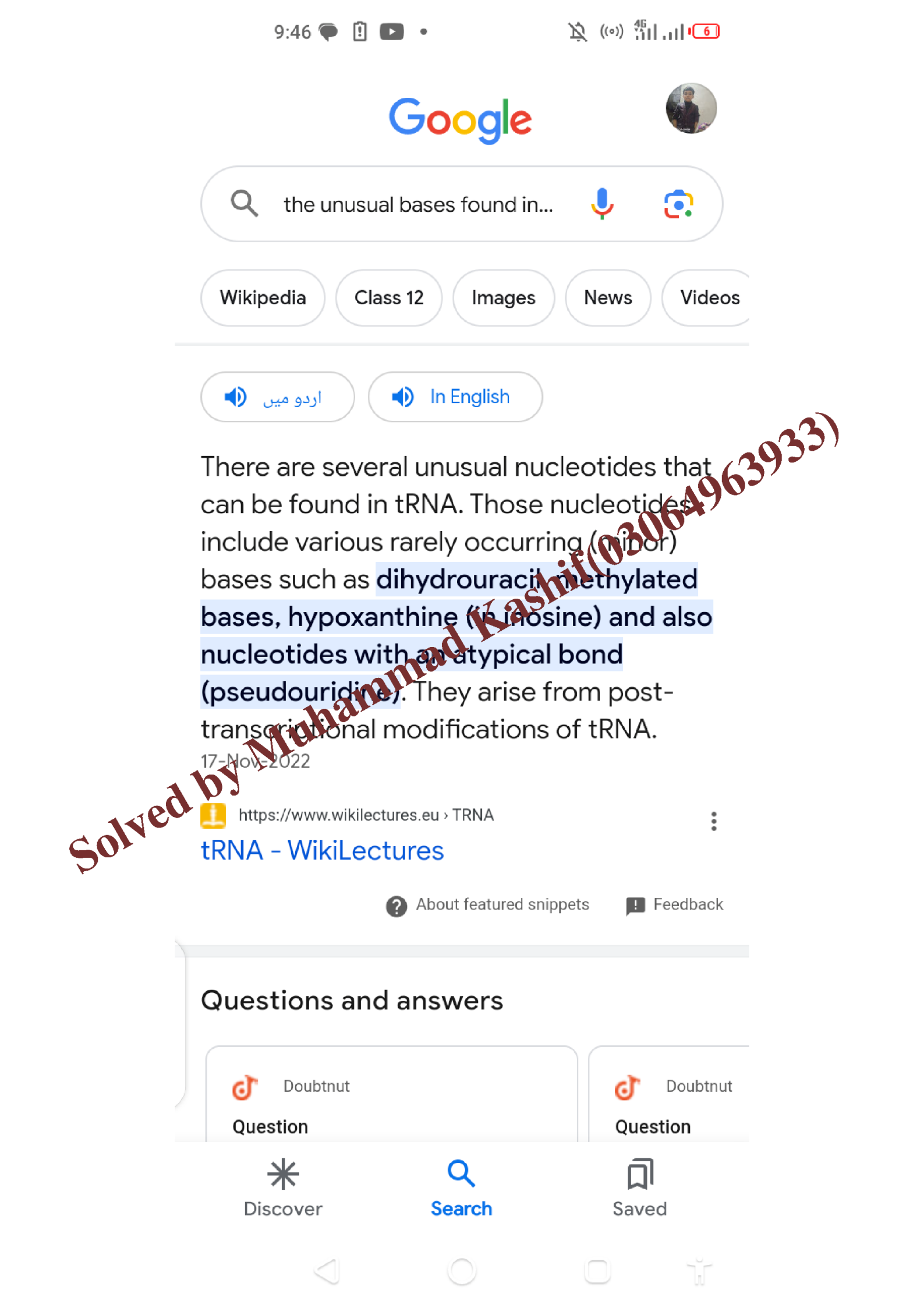
Select the correct option

○	1
○	2
●	3
○	4

Click to

Solved by Muhammad Kashif(03064963933)

Solved by Muhammad Kashif(03064963933)



9:45

ChatGPT
chat.openai.com

New chat

Model: GPT-4

AL

in eukaryotes translation is initiated by attachment of ribosomes to
5' Cap
Or
3'cap
Or
Tata box
Or
Poly A tail

In eukaryotes, translation is initiated by the attachment of ribosomes to the 5' cap of the mRNA molecule.

The 5' cap is a specially altered nucleotide on the 5' end of precurso messenger RNA and some other

Solved by Muhammad Kashif(03064963933)

Send a message

ChatGPT may produce inaccurate information about people, places, or facts. ChatGPT July 20 Version

9:35

vulms.vu.edu.pk/Qu

3

BIO302 - Molecular Biology (Quiz 2) Qui

Question # 10 of 10 (Start time: 09:35:15 PM, 25 July 2023)

Which molecular technique were used to find the direction of RNA editing by Kenneth Stuart and colleagues

Select the correct option

- ○ Hot Start PCR
- ○ PCR
- ○ Fast PCR
- ◉ RT-PCR

Click to Save

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9:35 vulms.vu.edu.pk/Qu 3

BIO302 - Molecular Biology (Quiz 2)

Question # 9 of 10 (Start time: 09:34:24 PM, 25 July 2023)

_____ number of domains are present in trigger factor

Select the correct option

- 2
- 6
- 3
- 5

Solved by Muhammad Kashif(03064963933)

9:34 vulms.vu.edu.pk/Qu 3

BIO302 - Molecular Biology (Quiz 2)

Question # 8 of 10 (Start time: 09:33:12 PM, 25 July 2023)

In eukaroytes, translation is initiated by binding of ribosomes to the _____

Select the correct option

- (●) 5' cap
- ( ) 3' cap
- ( ) Poly A tail
- ( ) TATA box

Solved by Muhammad Kashif(03064963933)

9:33

vulms.vu.edu.pk/Qu

BIO302 - Molecular Biology (Quiz 2)

Question # 7 of 10 (Start time: 09:32:25 PM, 25 July 2023)

One of the unusual bases found in tRNA is?

Select the correct option

- Adenine
- Cytosine
- Guanine
- Hypoxanthine

Solved by Muhammad Kashif(03064963933)

9:32 vulms.vu.edu.pk/Qu 3

BIO302 - Molecular Biology (Quiz 2)

Question # 6 of 10 (Start time: 09:32:08 PM, 25 July 2023)

Leader sequence of CRISPR array is A-T rich and _____ in lenght

Select the correct option

◉	500
○	200
○	190
○	370

Solved by Muhammad Kashif(03064963933)

9:30

vulms.vu.edu.pk/Qu

3

BIO302 - Molecular Biology (Quiz 2)

Question # 4 of 10 (Start time: 09:29:47 PM, 25 July 2023)

The ribosome must be able to bind at least_____tRNAs simultaneously to perform the peptidy

Select the correct option

- (●) 2
- ( ) 1
- ( ) 4
- ( ) 3

Click to S

Solved by Muhammad Kashif(03064963933)

9:29

vulms.vu.edu.pk/Qu

3

BIO302 - Molecular Biology (Quiz 2)

Question # 3 of 10 (Start time: 09:28:00 PM, 25 July 2023)

Paul C. Zamecnik and Mahlon B. Hoagland (1957) showed that:

Select the correct option

- Before their incorporation into DNA, amino acids are attached to a class of RNA molecules
- Before their incorporation into proteins, amino acids are attached to a class of DNA molecule
- Before their incorporation into proteins, nucleotides are attached to a class of RNA molecules
- Before their incorporation into proteins, amino acids are attached to a class of RNA molecule

Click

Solved by Muhammad Kashif(03064963933)

9:27 vulms.vu.edu.pk/Qu 3

BIO302 - Molecular Biology (Quiz 2)

Question # 2 of 10 (Start time: 09:27:41 PM, 25 July 2023)

Poly-A tail enhances mRNA level of translation by:

Select the correct option

○	Enhancing the activity of RNA polymerase
○	Silencing the recruitment of key translation initiation factors
○	Blocking the activity of DNA polymerase
●	Enhancing the recruitment of key translation initiation factors

Solved by Muhammad Kashif(03064963933)

9:27 vulms.vu.edu.pk/Qu 3

BIO302 - Molecular Biology (Quiz 2)

Question # 1 of 10 (Start time: 09:26:39 PM, 25 July 2023)

The Amino acids polymerizing machinery is composed of at least _____ RNA molecules.

Select the correct option

- 2
- 3
- 1
- 4

Cli

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8:51

Images Books News Shopping Videos

MORE RESULTS

What 3 things happen to pre-mRNA before it is sent out to the cytoplasm?

In eukaryotic cells, pre-mRNAs undergo three main processing steps: Capping at the 5' end. Addition of a poly(A) tail at the 3' end. Splicing to remove introns. 04-Aug-2021

https://bio.libretexts.org › Courses

2.2: RNA processing - Biology LibreTexts

MORE RESULTS

What is the processing of mRNA?

What are the 3 major steps involved in mRNA processing?

What carries mRNA to the cytoplasm?

How do mRNA copies of genes move from the nucleus to the cytoplasm?

What 3 modifications must happen before mRNA is sent out of the nucleus?

Solved by Muhammad Kashif(03064963933)

Discover Search Saved

8:41

vulms.vu.edu.pk/Qu

3

BIO302 - Molecular Biology (Quiz 2) Quiz St

Question # 10 of 10 (Start time: 08:40:34 PM, 25 July 2023)

The Amino acids polymerizing machinery is composed of at least _____ RNA molecules.

Select the correct option

- ○ 4
- ○ 2
- ○ 1
- ● 3

Click to Save An

Solved by Muhammad Kashif(03064963933)

8:40

vulms.vu.edu.pk/Qu

3

BIO302 - Molecular Biology (Quiz 2) Qu

Question # 9 of 10 (Start time: 08:39:01 PM, 25 July 2023)

_____ processes are carried out before mRNA is being exported to the cytoplasm from the nuc

Select the correct option

- (●) Transcription, Pre-mRNA splicing and Polyadenylation
- ( ) Transcription and polyadenylation
- ( ) Only Pre-mRNA splicing
- ( ) Transcription and Pre-mRNA splicing

Click to Save

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8:38 vulms.vu.edu.pk/Qu 3

BIO302 - Molecular Biology (Quiz 2)

Question # 8 of 10 (Start time: 08:38:08 PM, 25 July 2023)

The ribosome must be able to bind at least _____tRNAs simultaneously to perform the peptidy

Select the correct option

- 4
- 1
- 2
- 3

Click to S

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8:38 vulms.vu.edu.pk/Qu 3

BIO302 - Molecular Biology (Quiz 2)

Question # 7 of 10 (Start time: 08:37:04 PM, 25 July 2023)

The three nucleotide long units of protein coding region of the mRNA is called _____.

Select the correct option

- ○ Anticodon
- ◉ Codon
- ○ Pre-codon
- ○ Tricodon

Solved by Muhammad Kashif(03064963933)

8:37

vulms.vu.edu.pk/Qu

3

BIO302 - Molecular Biology (Quiz 2)

Question # 7 of 10 (Start time: 08:37:04 PM, 25 July 2023)

The three nucleotide long units of protein coding region of the mRNA is called _____.

Select the correct option

- ○ Anticodon
- ◉ Codon
- ○ Pre-codon
- ○ Tricodon

Solved by Muhammad Kashif(03064963933)

8:36 vulms.vu.edu.pk/Qu 3

BIO302 - Molecular Biology (Quiz 2)

Question # 6 of 10 (Start time: 08:36:47 PM, 25 July 2023)

Poly-A tail enhances mRNA level of translation by:

Select the correct option

○	Enhancing the activity of RNA polymerase
○	Blocking the activity of DNA polymerase
●	Enhancing the recruitment of key translation initiation factors
○	Silencing the recruitment of key translation initiation factors

Solved by Muhammad Kashif(03064963933)

8:36 vulms.vu.edu.pk/Qu 3

BIO302 - Molecular Biology (Quiz 2)

Question # 5 of 10 (Start time: 08:35:48 PM, 25 July 2023)

Leader sequence of CRISPR array is A-T rich and _____ in lenght

Select the correct option

- ○ 370
- ○ 200
- ◉ 500
- ○ 190

Solved by Muhammad Kashif(03064963933)

8:35 vulms.vu.edu.pk/Qu 3

BIO302 - Molecular Biology (Quiz 2)

Question # 4 of 10 (Start time: 08:35:28 PM, 25 July 2023)

During RNA splicing hnRNP family lack RS domain which causes suppression of RNA splicing. Why?

Select the correct option

- ○ Unable to bind RNA
- ○ Create steric blockage
- ○ Create steric hindrance
- ◉ Cannot recruit splicing machinery

Click

Solved by Muhammad Kashif(03064963933)

8:35

vulms.vu.edu.pk/Qu

3

BIO302 - Molecular Biology (Quiz 2)

Question # 3 of 10 (Start time: 08:34:48 PM, 25 July 2023)

New amino acid are added to the_____of the growing polypeptide chain during prot

Select the correct option

○	Amino terminus or carboxyl terminus
○	Amino terminus
●	Carboxyl terminus
○	Positively charged side

Click

Solved by Muhammad Kashif(03064963933)

8:34 • 141
vulms.vu.edu.pk/Qu 3

BIO302 - Molecular Biology (Quiz 2) Quiz Start Tir

Question # 2 of 10 (Start time: 08:33:17 PM, 25 July 2023)

Which molecular technique were used to find the direction of RNA editing by Kenneth Stuart and colleagues?

Select the correct option

- (●) RT-PCR
- ( ) Hot Start PCR
- ( ) Fast PCR
- ( ) PCR

Click to Save Answer &

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8:33 vulms.vu.edu.pk/Qu 3

BIO302 - Molecular Biology (Quiz 2)

Question # 1 of 10 (Start time: 08:33:04 PM, 25 July 2023)

_____ product of the following gene is only found in brain?

Select the correct option

○	ADAR4
○	ADAR2
○	ADAR1
●	ADAR3

Solved by Muhammad Kashif(03064963933)

8:23

Select text

- Humans and mice contain three ADAR genes: ADAR1, ADAR2, and ADAR3.

RNA Editing by Nucleotide Deamination

- The products of the first two are ubiquitous in the body, but the third gene product is found only in the brain.

Solved by Muhammad Kashif(03064963933)

RNA Editing by Nucleotide Deamination

- These enzymes are very specific. It would be disastrous if they deaminated every adenosine in an mRNA, so they select only certain adenosines in certain mRNAs.

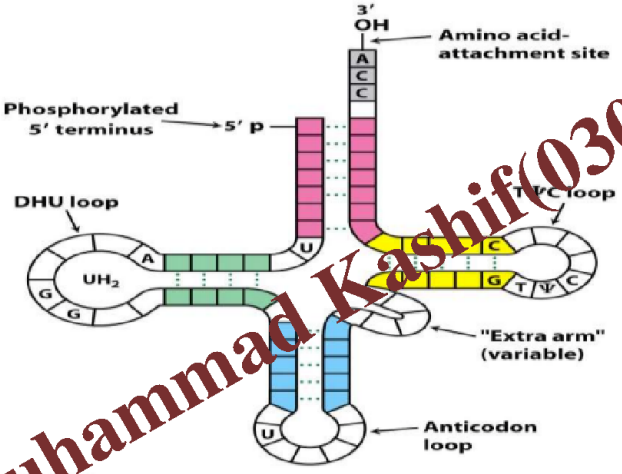
RNA Editing by Nucleotide Deamination

8:21 • 4G 17%

Select text

three stem-loops (referred to as the ψU loop, the D loop, and the anticodon loop), and a fourth variable loop.

Secondary Structure of tRNA



Solved by Muhammad Kashif(03064963933)

Secondary Structure of tRNA

- **The Acceptor Stem:**
- It is so-named because it is the site of attachment of the amino acid, is formed by pairing between the 5' and 3' ends the tRNA molecu

8:14

Select text

- It works by binding to sites near the 3' splice site of specific exons in those pre-mRNAs and recruiting the U2AF splicing factor.

Regulation by Activators and Repressors

- Most silencers are recognized by members of the heterogeneous nuclear ribonucleoprotein (hnRNP) family.
- These bind RNA but lack the RS domains and thus cannot recruit the splicing machinery.

Solved by Muhammad Kashif(03064963933)

Regulation by Activators and Repressors

- Instead, by blocking specific splice sites, they repress the use of those sites.
- In case of *Dscam*, for example, Hrp36 inhibits inclusion of exon 6 variants in the mRNA.

Regulation by Activators and Repressors

8:12

Select text

Secondary Structure of tRNA

- **The D Loop:**
- It takes its name from the characteristic presence of dihydrouridines in the loop.

Solved by Muhammad Kashif(03064963933)

Secondary Structure of tRNA

- **The Anticodon Loop:**
- As its name implies, contains the anticodon, a three-nucleotide-long sequence that is responsible for recognizing the codon by base pairing with the mRNA.

Secondary Structure of tRNA

- **The Variable Lo**
- It sits between th anticodon loop and the ψU loop and, as its name implies, varies in size from 3 to 21 bases

8:10

Select text

- A second feature that contributes to efficient translation is the presence of a poly-A tail at the extreme 3' end of the mRNA.
- This tail is added enzymatically by the enzyme poly-A polymerase.

Eukaryotic mRNAs

- Despite its location at the 3' end of the mRNA, the poly-A tail enhances the level of translation of them RNA by enhancing the recruitment of key translation initiation factors.

Solved by Muhammad Kashif(03064963933)

Eukaryotic mRNAs

- Importantly, in addition to their r in translation, the 5'- and 3'-end modifications als protect eukaryotic mRNAs from rapid degradation.

END

8:04 vulms.vu.edu.pk/Qu 3

BIO302 - Molecular Biology (Quiz 2)

Question # 10 of 10 (Start time: 08:02:29 PM, 25 July 2023)

Poly-A tail enhances mRNA level of translation by:

Select the correct option

○	Enhancing the activity of RNA polymerase
●	Enhancing the recruitment of key translation initiation factors
○	Silencing the recruitment of key translation initiation factors
○	Blocking the activity of DNA polymerase

Solved by Muhammad Kashif(03064963933)

8:02 vulms.vu.edu.pk/Qu 3

BIO302 - Molecular Biology (Quiz 2)

Question # 9 of 10 (Start time: 08:01:22 PM, 25 July 2023)

The anticodon loop contains _____.

Select the correct option

- (●) A three nucleotide anticodon
- ( ) A three nucleotide codon
- ( ) A four nucleotide codon
- ( ) A four nucleotide anticodon

Solved by Muhammad Kashif(03064963933)

8:01 vulms.vu.edu.pk/Qu 3

BIO302 - Molecular Biology (Quiz 2)

Question # 8 of 10 (Start time: 08:00:36 PM, 25 July 2023)

The three nucleotide long units of protein coding region of the mRNA is called _____.

Select the correct option

- ○ Tricodon
- ○ Anticodon
- ◉ Codon
- ○ Pre-codon

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8:00 vulms.vu.edu.pk/Qu 3

BIO302 - Molecular Biology (Quiz 2)

Question # 7 of 10 (Start time: 07:58:56 PM, 25 July 2023)

During RNA splicing hnRNP family lack RS domain which causes suppression of RNA splicing

Select the correct option

- ○ Create steric hindrance
- ○ Create steric blockage
- ○ Unable to bind RNA
- ◉ Cannot recruit splicing machinery

Solved by Muhammad Kashif(03064963933)

7:56

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3

BIO302 - Molecular Biology (Quiz 2)

Question # 5 of 10 (Start time: 07:56:24 PM, 25 July 2023)

The Amino acids polymerizing machinery is composed of at least _____ RNA molecules.

Select the correct option

- (●) 3
- ( ) 1
- ( ) 4
- ( ) 2

Click to S

Solved by Muhammad Kashif(03064963933)

7:56 vulms.vu.edu.pk/Qu 3

BIO302 - Molecular Biology (Quiz 2)

Question # 4 of 10 (Start time: 07:54:48 PM, 25 July 2023)

_____ product of the following gene is only found in brain?

Select the correct option

●	ADAR3
○	ADAR1
○	ADAR4
○	ADAR2

Solved by Muhammad Kashif(03064963933)

7:54 • 4G 24

vulms.vu.edu.pk/Qu 3

BIO302 - Molecular Biology (Quiz 2)

Question # 3 of 10 (Start time: 07:54:01 PM, 25 July 2023)

The exit tunel in bacterial ribosome is about _____ long

Select the correct option

○	50-80A^0
●	80-100 A^0
○	10-3- A^0
○	5-18 A^0

Solved by Muhammad Kashif(03064963933)

7:53

vulms.vu.edu.pk/Qu

3

BIO302 - Molecular Biology (Quiz 2) Quiz Start

Question # 2 of 10 (Start time: 07:52:28 PM, 25 July 2023)

How many enzymatic steps are required by all aminoacyl-tRNA synthetases to attach an amino acid to a tRNA?

Select the correct option

- 1
- 4
- 3
- 2 (selected)

Click to Save Answe

Solved by Muhammad Kashif(03064963933)

7:52

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3

BIO302 - Molecular Biology (Quiz 2) Quiz Sta

Question # 1 of 10 (Start time: 07:51:10 PM, 25 July 2023)

The transcription machinery and the translation machinery in _____ are located in the same compartment.

Select the correct option

- ○ Eukaryotes
- ○ Viruses
- ○ Fungi
- ◉ Prokaryotes

Click to Save Answ

Solved by Muhammad Kashif(03064963933)